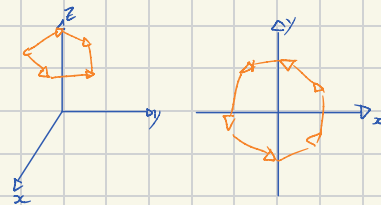


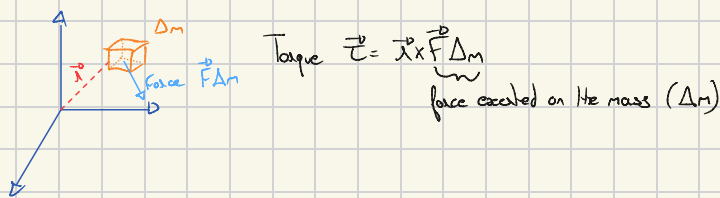
Lecture 33: Topological considerations; Maxwell's equations

CURL: recall: $\vec{v}$ velocity field $\Rightarrow \text{curl}(\vec{v})$ measures $2 \times$ angular velocity vector (rotation part of the motion)

E.g.: uniform rotation around z -axis
 $\vec{v} = \omega(-y\hat{i} + x\hat{j})$
 $\nabla \times \vec{v} = 2\omega\hat{k}$



For a force field?



For translation: $\text{Force} = \text{Acceleration} = \frac{d(\text{Velocity})}{dt}$
 Mass

For rotation: $\frac{\text{Torque}}{\text{Moment of inertia}} = \text{Angular acceleration} = \frac{d(\text{Angular Velocity})}{dt}$
 curl

Consequence: if force $\vec{F}$ derives from a potential $\Rightarrow \text{curl } \vec{F} = 0 \Rightarrow \vec{F}$ does not induce any rotation motion.

Electric/magnetic fields: Maxwell's equations
 $\vec{E} \quad \vec{F} = q\vec{E}$
 $\vec{B} \quad \vec{F} = q\vec{v} \times \vec{B}$

Div and curl of $\vec{E}$

* Gauss-Coulomb law: $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$ electric charge density

$$\oint_S \vec{E} \cdot d\vec{S} = \iiint_D \text{div } \vec{E} dV = \frac{1}{\epsilon_0} \iiint_D \rho dV = \frac{Q}{\epsilon_0} \rightarrow \text{Total electric charge in } D.$$

* Faraday's law: $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Stokes

$$\oint_C \vec{E} \cdot d\vec{r} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{S} = \iint_S \left(-\frac{\partial \vec{B}}{\partial t}\right) \cdot d\vec{S}$$

Remark: $\nabla \cdot \vec{B} = 0$

$$\nabla \cdot \vec{B} = \mu_0 \underbrace{\vec{J}}_{\text{vector current density}} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$